CONTINUĀ

Predictive Network Continuity

The network changes. The session does not.

ADVANCED TECHNOLOGY RESEARCH COUNCIL

EDGE

ACCESS
4
wired to satellite

TRAFFIC
5
mission classes

GOAL
1
continuous session

PROPOSAL DECK

by Team Kanban

Advanced Technology Pioneers 2026

EXECUTIVE SUMMARY

One session. Four networks. Zero operator disruption.

CONTINUA keeps critical applications usable as a mobile user transitions across heterogeneous access networks.

How might we keep a user's experience unbroken as they move between Wi-Fi, mobile, satellite and wired networks?

CONTINUA in one sentence

An application-aware continuity layer that predicts link degradation, prepares the next path before failure, and protects critical traffic during the handoff.

PREDICT | PRE-WARM | PRIORITIZE

Prototype

agent + gateway + dashboard

Simulation

repeatable multi-access scenarios

Predictive model

short-horizon QoE risk

Comparative results

baselines + application metrics

Team Kanban combines product, AI, networking and experimental rigor.

Awaiz Ahmed

Technical & Product Lead

BSc Software Engineering + Software Development student

CORE CAPABILITIES

- AI/ML & QoE prediction
- Python & Linux
- ROS & robotics integration
- FastAPI & Next.js
- Docker & cloud-native delivery
- Network engineering

Leads architecture, model, integration, dashboard and pitch.

Simulation & Experiment Lead

Mininet-WiFi, tc netem, automation, statistics

Mahra Alshamsi

Network & 5G Lead

MPTCP, QoS, Linux routing, wireless systems

Mohammad Umar

Research & Experience Lead

literature, usability, documentation, demo narrative

Obaid Mukkadam

PROBLEM & USE CASE

The handoff is brief. The mission impact is not.

Use case: a mobile response or inspection platform moves from a command facility into remote terrain.

Operator should never need to reconnect or regain control

Mission traffic is not equal

Command & safety	CRITICAL
Telemetry	CRITICAL
Live video	IMPORTANT
Voice	IMPORTANT
Logs & files	OPPORTUNISTIC

Session reset
operator re-authenticates or restarts the application

Command delay
control messages miss operational deadlines

Video freeze
situational awareness drops during a transition

Telemetry gaps
the latest state becomes stale or incomplete

RESEARCH & GAP

Connectivity mechanisms exist. Mission-aware continuity is still missing.

CONTINUA adapts standards-based multipath concepts to predictive, application-aware field operations.

BASELINE 0

Single-path routing

WHAT IT GIVES US

Simple and universally available.

WHAT IS STILL MISSING

Route or source-address changes can interrupt active sessions; switching usually begins after degradation is visible.

BASELINE 1

Multipath TCP

WHAT IT GIVES US

One reliable byte stream can use multiple subflows and fail over between paths.

WHAT IS STILL MISSING

The transport layer alone does not know which application is mission-critical or what future path risk looks like.

REFERENCE

Steering / redundancy

WHAT IT GIVES US

Traffic can be steered, switched, split or duplicated across accesses.

WHAT IS STILL MISSING

Full deployment can be complex, while always-on duplication can waste scarce or costly bandwidth.

OUR ADAPTATION

CONTINUA

WHAT IT GIVES US

Predicts risk, pre-warms the next path and applies a different policy to each traffic class.

WHY IT FITS

Built on open standards and testable in software emulation before field deployment.

Standards-aligned foundation: IETF RFC 8684 (MPTCP) | Linux MPTCP | 3GPP TS 24.193 (ATSSS)

PROPOSED SOLUTION

CONTINUA turns live telemetry into proactive handoff decisions.

A closed-loop controller observes every link, predicts near-term QoE risk and takes the least disruptive action.

1. **Observe** — collect live link and motion signals
2. **Predict** — estimate QoE violation in 3-5 seconds
3. **Prepare** — pre-warm the best backup path
4. **Steer** — switch, split or briefly duplicate
5. **Explain** — log the reason and confidence

INPUTS

RTT | jitter | packet loss | throughput
Wi-Fi RSSI | interface state | motion | cost

ACTIONS

Pre-warm backup | switch primary | split traffic
briefly duplicate critical packets | adapt video |
pause logs

PATH SCORE = QoE FIT - PREDICTED RISK - COST - HANDOFF PENALTY

COMMAND	TELEMETRY	VIDEO	LOGS
lowest latency + brief duplication	freshest-data policy	predictive switch + adaptive bitrate	cheapest / highest throughput

A deployable continuity layer - not a replacement for existing networks.

The first prototype runs entirely in software and can later map onto physical routers, gateways or mobile platforms.

CONTROL PLANE

Link Monitor → QoE Predictor → Policy Engine → Decision Log

No specialized hardware required

Mobile platform

Control
Telemetry
Video
Voice
Logs

CONTINUA agent

Flow classifier
MPTCP / UDP relay
Local policy executor

Multi-access paths

Wired
Wi-Fi
5G
Satellite

Continuity gateway

Stable endpoint
Encrypted tunnel
Duplicate suppression
Session state

Operator

Console
Live feed
Alerts

Authenticated peers

Encrypted transport

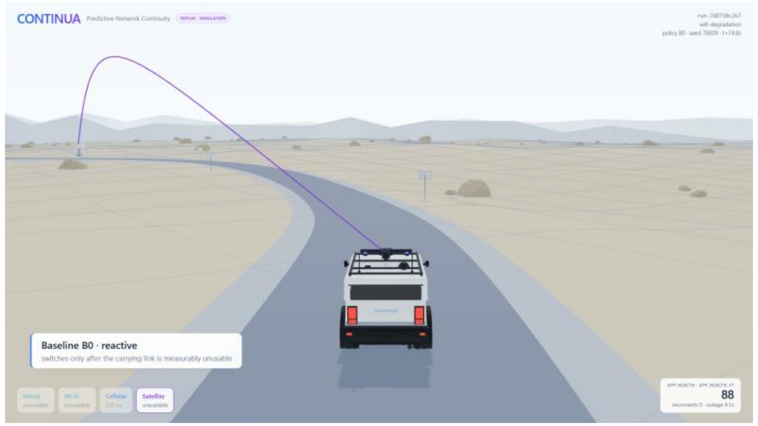
Least-privilege agent

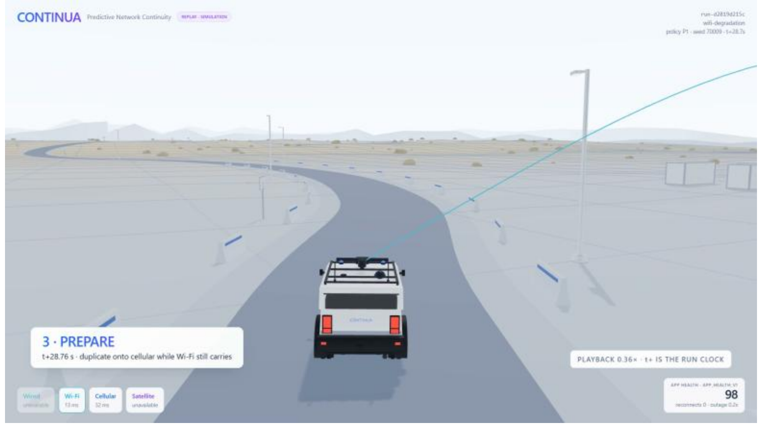
Auditable decisions

WORKING PROTOTYPE

A working prototype: visible, stressable and measured.

The dashboard visualizes logged handoff events, application health and the reasoning behind every network decision. Live at continua.kanbanstudios.ae





01 **Live multi-link state**

Wired, Wi-Fi, 5G and satellite visible together.

02 **Predictive handoff**

The next path is prepared before the current link fails.

03 **Application-aware QoS**

Latency, loss, throughput and service health stay visible.

04 **Operator mission view**

Live feed, vehicle state and transition status in one place.

Evidence first: repeatable emulation, clear baselines, application-level QoE.

Because no sponsor dataset is provided, every assumption, network profile and workload is explicitly documented.

ASSUMPTIONS

- No live operator network or private sponsor data is required.
- Four access types are represented by controlled Linux profiles.
- Identical mobility and workload traces are used for each mechanism.
- Dashboard values become evidence only after logged experiment runs.

PROTOTYPE STACK

- Linux MPTCP and network namespaces
- Mininet-WiFi for movement and AP handovers
- tc netem for delay, loss, jitter and rate profiles
- Python, scikit-learn, FastAPI, Next.js and WebSockets

EXPERIMENTAL DESIGN

- B0: single-path TCP with route switching
- B1: reactive / default MPTCP
- P1: CONTINUA predictive policy
- 20+ repeated runs per core scenario, fixed seeds and confidence intervals

SCENARIOS

Gradual Wi-Fi fade · Sudden link outage · 5G congestion · Satellite fallback · Flapping link

METRICS

interruption time · p95 command latency · deadline misses · video stall time · telemetry age · satellite MB · unnecessary handoffs

Simulation results are measured and reproducible. Linux MPTCP and Mininet-WiFi emulation remain planned; they are not verified on the current host.

IMPACT & ROADMAP

From working prototype to field-ready continuity.

Success means better mission experience, smarter use of scarce links and evidence that can be reproduced.

Mission continuity

Keep critical control, telemetry and video usable as the access network changes.

Efficient bandwidth

Use satellite and duplicated traffic only when risk and application priority justify it.

Explainable automation

Show operators what changed, why a handoff occurred and how confident the system was.

TARGET

Zero application reconnects in supported handoffs

BUILD ROADMAP - COMPLETE

- 01 Baseline & emulator
- 02 MPTCP & gateway
- 03 Predictor & policy
- 04 Dashboard integration
- 05 Experiments & video

SUCCESS DEFINITION

- Lower interruption and command deadline misses than baselines
- Lower satellite use than always-on redundancy

AI USE DECLARATION

AI tools support ideation. The team does coding, documentation and validates every architecture decision, experiment, result and conclusion.

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